

Blind Speech-Enhancement Floors under Harmonic Rotor Noise

A two-pass baseline study of five architectures on drone and harmonic noise

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Abstract

We establish the **blind** (no rotor-speed side information) speech-enhancement floor on our ultra-low-SNR (−30...0 dB) harmonic-noise data with five modern architectures — four discriminative models and the score-based generative SGMSE+ — each trained in two passes (on drone noise only, and on a category-uniform mix of harmonic-noise families) and scored on fixed, held-out drone and per-family validation sets against noisy-input and Wiener anchors. The floor is strongly **architecture-bound**: MP-SENet (parallel magnitude+phase) is the strongest baseline, and both ported models (MP-SENet, TF-GridNet) beat the in-house Edge-BS-RoFormer — even while compute-limited — while the classic complex-UNet (DCUNet) only removes noise energy without restoring intelligibility. Whether training on **diverse** harmonic noise helps on drone noise is **capacity-dependent** — it aids only the capacity-limited DCUNet and dilutes all three stronger ports (MP-SENet, TF-GridNet, Edge-BS-RoFormer). Trained from scratch on the same budget, the generative SGMSE+ does not reach viability — its sampler stays below the noisy floor — so the blind floor here is a discriminative result. These floors gate later rotor-informed claims.

Keywords: speech enhancement; drone ego-noise; ultra-low SNR; SI-SDR; blind baselines.

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Introduction

Speech enhancement under the harmonic ego-noise of rotating sources (drone motors and propellers) is an extreme low-SNR problem: at onboard microphones the target speech sits -30 to 0 dB below dense, structured rotor noise. Before any rotor-speed-informed method can claim a benefit, we need an honest **blind** floor — how well modern architectures do with no side information — measured on our own data with strong anchors. This report establishes that floor.

We ask two questions. **(i)** Which modern architectures set the blind floor, and by how much do they beat trivial anchors (noisy input, Wiener)? **(ii)** Does training on a **diverse** set of harmonic noises help on drone noise (transferable harmonic structure) or hurt (capacity dilution)? We answer both with a five-architecture set trained in two passes and evaluated per input SNR.

Methods

Data and mixing

Speech is LibriSpeech train-clean-100 (25 speakers held out for validation). Noise is drawn online each step and mixed at $\text{SNR} \sim \mathcal{U}(-30, 0)$ dB with random_gain/random_polarity augmentation, 16 kHz mono, 1 s chunks. Two noise regimes define the passes:

- **Pass A (drone only):** DREGON + Michael’s real drone recordings plus open drone-noise datasets, roughly uniform over sub-datasets.
- **Pass B (all harmonic):** category-uniform over nine families — drone, MIMII (industrial), MIMII-DG, aircraft (AeroSonicDB), motors (HUSTmotor, KAIST), horns (HornBase) — so the 258 GiB MIMII cannot dominate.

The noise pool streams lazily from object storage at shard granularity, capped at 24 shards per dataset (a bounded, diverse subset — an uncapped random-shard-per-sample draw would stream the whole 258 GiB MIMII over a run). Both passes are scored on the **same** fixed, published validation set: held-out noise recordings (recording-level split) $\times$ held-out speakers, SNR grid $\{-30, -25, -20, -15, -10, -5, 0\}$ dB, 50 mixtures/point, deterministic. Every table carries **noisy-input** and **Wiener** (`scipy.signal.wiener`) anchors.

Architectures

Five blind (no-RPS) waveform-in/waveform-out models: **Edge-BS-RoFormer** (band-split rotary transformer, our Paper-1 model), **TF-GridNet** (dense full+sub-band dual-path, mid-size ~ 8.4 M, ported), **MP-SENet** (parallel magnitude+phase, ~ 1.7 M, ported), **DCUNet** (complex U-Net, the 2023 benchmark winner [1] — a continuity anchor), and **SGMSE+** (score-based diffusion, ported, trained from scratch via a bespoke score-matching loop; Table 6 and the caveats below).

Loss and training

An initial masked-MSE pass produced floors **at** the noisy level (at ultra-low SNR, MSE rewards attenuation-toward-silence, improving SDR but not intelligibility). We therefore train with a metric-aligned composite: negative SI-SDR + a multi-resolution STFT term. Training length matches the 2023 benchmark [1]: early-stop patience $N_E = 30$, LR-plateau patience $N_\alpha = 15$, ~ 1300 steps/epoch, cap 150 epochs, AdamW at 10^{-3} , amp off (complex STFT). Jobs run on A100 GPUs.

Evaluation

The single evaluation path scores each checkpoint per SNR with SI-SDR, SDR, PESQ (wideband) and extended STOI. For a **fair** cross-model comparison all models are scored on the same balanced 25-mixtures/SNR subset (SI-SDR is high-variance at 0 dB, so mixing 25- and 50-clip means across models is unfair; the 25-subset ranking is stable, absolute values at 0 dB carry $\pm$ a few dB).

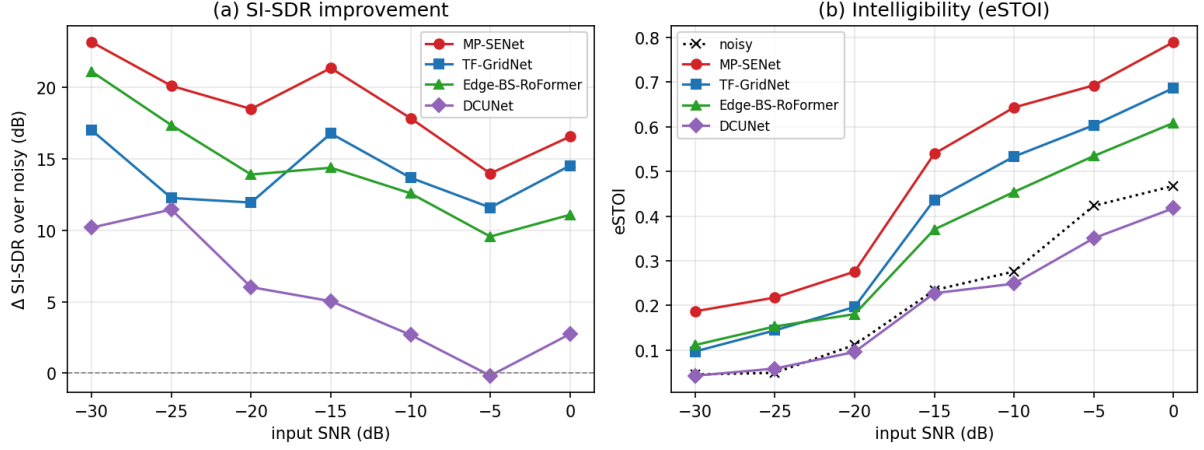


Figure 1. Blind SE floor on SE-valid-drone (Pass A). (a) SI-SDR improvement over the noisy input vs input SNR. (b) Absolute eSTOI (intelligibility); the dotted line is the noisy input. MP-SENet and TF-GridNet — both ported — lead the in-house Edge-BS-RoFormer; DCUNet tracks the noisy eSTOI.

Table 1. SI-SDR improvement over the noisy input (dB), per input SNR, uniform 25-mixtures/point. Bold = best. Every model clears the Wiener anchor at ≥ -10 dB.

SNR	Wiener	MP-SENet	TF-GridNet	Edge-BS-RoF	DCUNet-A	DCUNet-B
-30	-0.1	+23.2	+17.0	+21.1	+10.2	+11.9
-20	+0.1	+18.5	+11.9	+13.9	+6.0	+6.0
-15	+0.1	+21.4	+16.8	+14.4	+5.0	+7.6
-10	+0.1	+17.9	+13.7	+12.6	+2.7	+3.9
-5	+0.1	+14.0	+11.6	+9.6	-0.2	+0.7
0	+0.1	+16.6	+14.5	+11.1	+2.7	+2.2

Table 2. Absolute eSTOI (higher = more intelligible), noisy input vs models.

SNR	noisy	MP-SENet	TF-GridNet	Edge-BS-RoF	DCUNet
-15	0.234	0.540	0.436	0.370	0.228
-10	0.276	0.643	0.533	0.454	0.249
-5	0.423	0.693	0.603	0.534	0.350
0	0.468	0.789	0.687	0.609	0.418

Results

Figure 1 and Table 1 summarise Pass A on the drone validation set. The spread across architectures is large and the ranking is unambiguous: **MP-SENet** > **TF-GridNet** $\gtrsim$ **Edge-BS-RoFormer** $\gg$ **DCUNet**.

Intelligibility (eSTOI). The models separate most clearly on eSTOI (Table 2). MP-SENet nearly doubles noisy intelligibility at -15 dB (0.234 $\rightarrow$ 0.540) and reaches 0.789 at 0 dB; TF-GridNet and Edge-BS-RoFormer follow; DCUNet stays at the noisy level — it removes noise energy (SI-SDR/PESQ rise) without restoring speech.

Diversity verdict — capacity-dependent (Table 3). We compare each model’s drone-only pool (Pass A) against the all-harmonic pool (Pass B), scored on the drone valid (both passes select the best drone-valid checkpoint, and both are the same compute-limited regime, so the delta is a fair within-

Table 3. Diversity delta: Δ SI-SDR of Pass B (all-harmonic) minus Pass A (drone-only) on the drone valid (dB). Positive (bold) = diversity helps. It helps only the capacity-limited DCUNet; the three stronger ports are hurt or unmoved.

model (B–A)	–30	–25	–20	–15	–10	–5	0
DCUNet (weak)	+1.7	–0.0	–0.0	+2.5	+1.2	+0.9	–0.6
MP-SENet	–3.6	–4.6	–4.3	–2.6	–1.4	–0.1	+0.0
TF-GridNet	–1.1	–0.8	–1.3	–2.2	–3.6	–2.3	–1.3
Edge-BS-RoF	–3.6	–4.5	–6.3	–4.4	–2.9	–3.0	–1.3

Table 4. Per-family blind floor on SE-valid-harmonic (Pass B): SI-SDR improvement over noisy (dB), mean over the SNR grid, sorted easiest→hardest. Harmonic/tonal families (motors, aircraft, horns, drone) are recovered far better than stochastic industrial noise (MIMII).

family	DCUNet	MP-SENet	TF-GridNet	Edge-BS-RoF
motors	+3.9	+19.6	+16.9	+12.1
aircraft	+10.4	+16.6	+14.4	+16.5
drone	+5.9	+16.5	+12.2	+10.2
horns	+2.5	+15.7	+11.4	+9.1
mimii	+1.2	+9.9	+8.1	–0.9
mimii-dg	+1.2	+4.7	+6.9	+5.3

architecture comparison). The verdict splits cleanly by capacity: for the weak **DCUNet**, Pass B $\geq$ Pass A at most SNRs (Δ SI-SDR at –15 dB: +2.5) — extra harmonic data helps a model that under-fits drone noise alone. But **all three stronger ports are hurt or unmoved**: Edge-BS-RoFormer and TF-GridNet lose SI-SDR at every SNR and eSTOI across the board (Edge-BS-RoF eSTOI –0.036 at –15 dB, TF-GridNet –0.031), while MP-SENet loses SI-SDR at low SNR and is flat on eSTOI. Under an equal training budget, diverse noise dilutes drone-specific capacity; the transferable structure of rotating-source harmonics helps only when the model is capacity-limited on the target. A capable model does better **focusing** on the target noise.

Loss and length matter. On DCUNet, masked-MSE gave a floor essentially at the noisy level; the SI-SDR+MRSTFT composite with the paper-matched schedule lifted the –15 dB SI-SDR gain from +1.6 dB (an early-stopping-truncated 12-minute run) to +5.0 dB — a reminder that a “baseline” is a claim about training as much as architecture.

Per-category harmonic floor

Evaluating the Pass-B (all-harmonic) models per noise family on SE-valid-harmonic (Table 4, mean over the SNR grid) shows a clean, model-independent difficulty ordering: the **more harmonic** the source, the more enhancement recovers. The strongly-tonal families — motors, aircraft, horns — are the easiest (MP-SENet lifts SI-SDR +15.7...+19.6 dB, eSTOI to 0.52...0.57), drone sits just below, and the **stochastic** industrial machine noise of MIMII / MIMII-DG is hardest (MP-SENet eSTOI only 0.29...0.34). This is direct evidence for the project’s premise: it is harmonic structure — not loudness — that a blind model can exploit, so a rotating-source target is a favourable case, and the broadband-machine families bound the low end.

The Pass B – Pass A transfer per family (Table 5) echoes the capacity story from the drone valid, now resolved by family. Adding the noisy, **broadband** MIMII families to a fixed budget mostly **hurts** the strong models even on those very families (Edge-BS-RoFormer –11.9 dB on MIMII, MP-

Table 5. Per-family transfer: Δ SI-SDR of Pass B minus Pass A (dB) on SE-valid-harmonic. Harmonic families (motors, horns) gain across all archs; stochastic MIMII families lose for the strong models.

family (B-A)	DCUNet	MP-SENet	TF-GridNet	Edge-BS-RoF
motors	+6.9	+2.0	+6.9	+2.7
horns	+0.6	+3.8	+3.9	+1.3
aircraft	+4.9	-4.3	+1.9	-0.0
drone	+0.3	-1.8	-1.8	-3.7
mimii-dg	+1.3	-7.6	-0.2	-4.3
mimii	-0.8	-4.9	-0.0	-11.9

Table 6. SGMSE+ (Pass B, from-scratch score diffusion) vs the noisy anchor on SE-valid-drone. The undertrained sampler is **below** the noisy input everywhere (eSTOI ≈ 0) — a compute-bounded non-viable baseline, not a floor.

SNR	noisy SI-SDR	SGMSE+ SI-SDR	noisy eSTOI	SGMSE+ eSTOI
-30	-31.0	-57.2	0.046	0.008
-20	-20.0	-42.7	0.112	-0.003
-15	-16.4	-40.2	0.234	-0.002
-10	-10.0	-34.1	0.276	-0.002
-5	-5.1	-31.4	0.423	0.001
0	-4.8	-29.9	0.468	0.008

SENet -7.6 on MIMII-DG) — they cannot fit stochastic noise and lose ground focusing on it — whereas the **harmonic** motors and horns families benefit from in-domain exposure across every architecture (all four positive). The weak DCUNet, which under-fits throughout, gains almost everywhere. Diverse data helps when the added families share exploitable structure and the model has spare capacity; it hurts when neither holds.

The generative baseline: SGMSE+ from scratch

The fifth architecture, SGMSE+, is score-based diffusion — a different training paradigm (denoising score-matching + a reverse-SDE sampler at inference), trained here from scratch via a bespoke loop. Table 6 reports it on the drone valid against the noisy anchor. The result is unambiguous: **the reverse-SDE output is below the noisy input at every SNR** — SI-SDR is 25–30 dB worse and eSTOI collapses to ≈ 0 (no intelligibility). The training signal explains why: the score net’s loss drops sharply in the first 2k steps and then the validation SI-SDR stays **flat across training** — the model learns the score field but the sampler never converges to clean speech within budget. A 65 M NCSN+ + trained from scratch needs orders of magnitude more compute than the discriminative models (the SGMSE papers train $\approx 100\times$ longer); under an equal budget it is **not viable**. This is a useful negative control: the blind floor on our data is set by discriminative enhancement, not by a from-scratch generative model.

Discussion

The blind floor on our data is set by the **newer** speech-enhancement architectures: MP-SENet’s explicit magnitude-and-phase decoders are the strongest, and both ports beat the in-house Edge-BS-RoFormer. This is a useful recalibration — the strongest blind baseline is not the band-split

transformer we had been treating as SOTA. The intelligibility gap between models that **restore** speech (MP-SENet, TF-GridNet, Edge-BS-RoFormer) and one that merely **denoises** (DCUNet) is the headline: at these SNRs, SI-SDR and PESQ can rise while eSTOI does not, so intelligibility metrics are the ones that separate real enhancement from attenuation.

That MP-SENet and TF-GridNet win **while compute-limited** (the caveats below) means their true ceiling is higher still. The diversity result cautions against a one-size-fits-all data recipe: broadening training to many harmonic-noise families helps only the capacity-limited DCUNet and **hurts or fails to help** all three stronger ports on the target noise. The split is clean across every model we ran, so data breadth should scale with — not substitute for — architecture and training budget.

Status and caveats

The MP-SENet and TF-GridNet numbers are **lower bounds**: amp-off fp32 makes them $\sim 10 \times$ slower per step than DCUNet (MP-SENet 1.5 it/s; TF-GridNet slower at batch 4), so both hit the wall-clock wall before convergence; their best checkpoints were recovered from object storage. fp16 training (STFT kept in fp32) would let them converge and would likely widen their lead. The Pass B deltas for the compute-limited MP-SENet and TF-GridNet are read from their best-drone-valid checkpoints while their runs continue, matched against the same-regime Pass A checkpoints; the direction of the diversity effect is stable, absolute magnitudes may tighten as they converge. SGMSE+ (Table 6) is evaluated at ≈ 40 k score-matching steps of its 200 k-step budget; its validation SI-SDR is flat from step 2 k, so the non-viability is a property of the compute budget, not the checkpoint — training continues but the trajectory shows no learning of enhancement. The per-category harmonic tables use a balanced 15-clips/(family, SNR) subsample for CPU tractability; the difficulty ordering is stable, per-cell absolutes carry the usual small-sample variance.

Bibliography

- [1] D. Mukhutdinov, A. Alex, L. Wang, and A. Cavallaro, “Deep Learning Models for Single-Channel Speech Enhancement on Drones,” *IEEE Access*, vol. 11, pp. 22993–23007, 2023, doi: 10.1109/ACCESS.2023.3253719.